

1.1 THE THREE I's OF VIRTUAL REALITY

It is clear from the foregoing description that virtual reality is both interactive and immersive. These features are the two I's that most people are familiar with. There is, however, a third feature of virtual reality that fewer people are aware of. Virtual reality is not just a medium or a high-end user interface, it also has applications that involve solutions to real problems in engineering, medicine, the military, etc. These applications are designed by virtual reality developers. The extent to which an application is able to solve a particular problem, that is, the extent to which a simulation performs well, depends therefore very much on the human *imagination*, the third "I" of VR. Virtual reality is therefore an integrated trio of immersion–interaction–imagination, as shown in Figure 1.2. The imagination part of VR refers also to the mind's capacity to perceive nonexistent things. The triangle in Figure 1.2, for example, is easily "seen" by the reader, yet it only exists in his or her imagination.

1.2 A SHORT HISTORY OF EARLY VIRTUAL REALITY

Virtual reality is not a new invention, but dates back more than 40 years. In 1962, U.S. Patent #3,050,870 was issued to Morton Heilig for his invention entitled *Sensorama Simulator*, which was the first virtual reality video arcade. As shown in Figure 1.3, this early virtual reality workstation had three-dimensional (3D) video feedback (obtained with a pair of side-by-side 35-mm cameras), motion, color, stereo sound, aromas, wind effects (using small fans placed near the user's head), and a seat that vibrated. It was thus possible to simulate a motorcycle ride through New York, where the "rider" sensed the wind and felt the pot-holes of the road as the seat vibrated. The rider could even smell food when passing by a store.

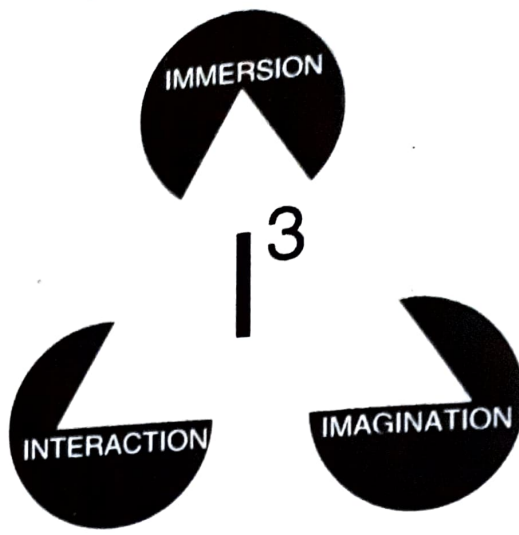


Fig. 1.2 The three I's of virtual reality, immersion–interaction–imagination. Adapted from Burdea [1993]. Reprinted by permission.

Heilig also realized the possibilities of head-mounted television. He designed a simulation mask that included 3D slides with wide peripheral effects, focusing controls and optics, stereophonic sound, and the capability to include smell. A depiction from U.S. Patent #2,955,156 issued to him on October 4, 1960, is shown in Figure 1.4. Heilig, a cinematographer by profession, was well ahead of his time. Like Jules Verne, he imagined a new machine that would replace the classical cinematographic experience of today. He was also like Thomas Edison, an inventor who not only dreamed ideas, but also transformed them into real machines. At the time of Heilig's inventions, nobody realized the revolutionary technological progress they represented.

Heilig's initial work on head-mounted displays (HMD) was continued by Ivan Sutherland. In 1966 Sutherland used two cathode ray tubes (CRTs) mounted along the user's ears. Today's high-end HMDs use miniature CRTs mounted in the same configuration. Since the tubes available in 1966 were much heavier than those in use today, Sutherland had to rely on a mechanical arm to support the weight of the display. This mechanical arm had potentiometers that measured the user's view direction. Most of today's HMDs use noncontact position tracking (magnetic or ultrasound), but this technology was not available in the 1960s.

While working on his head-mounted display, Sutherland realized that he could use computer-generated scenes instead of analog images taken by cameras, and began to design such a scene generator. This was the precursor of the modern graphics accelerator, a key part of VR hardware. The computer generates a sequence of scenes each a bit different and each displayed in a fraction of a second. The overall effect is that of animations such as flyby simulations used to train pilots. Early graphics scene generators produced by Evans and Sutherland around 1973 could display simple scenes of only 200–400 polygons. Each scene took about 1/20 sec to compute and display, so that about 20 scenes (or frames) were displayed every second. More complex scenes were made of more polygons, took a longer time to display, and therefore consisted of fewer frames per second. As a consequence, animation (which requires more than 16 frames/sec to be smooth) suffered.

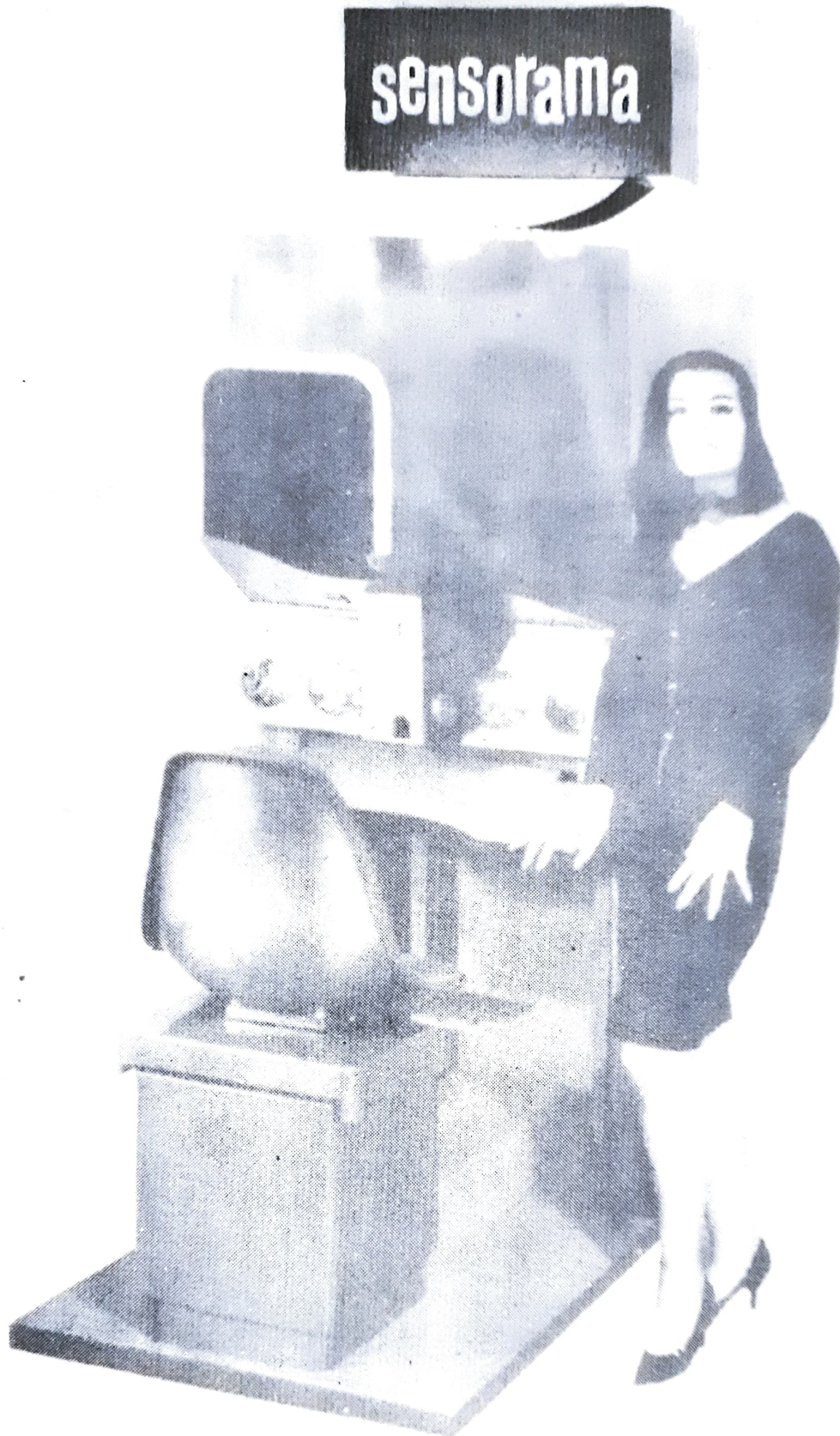


Fig. 1.3 The Sensorama Simulator prototype. Courtesy of M. Heilig.

Sutherland's vision of an "ultimate display" to the virtual world was not limited to graphics. In 1965 he predicted that the sense of touch (or haptics) would be added in order to allow users to feel the virtual objects they saw [Sutherland, 1965]. This idea was made reality by Frederick Brooks, Jr., and his colleagues at the University of North Carolina at Chapel Hill. By 1971 these scientists demonstrated the ability to

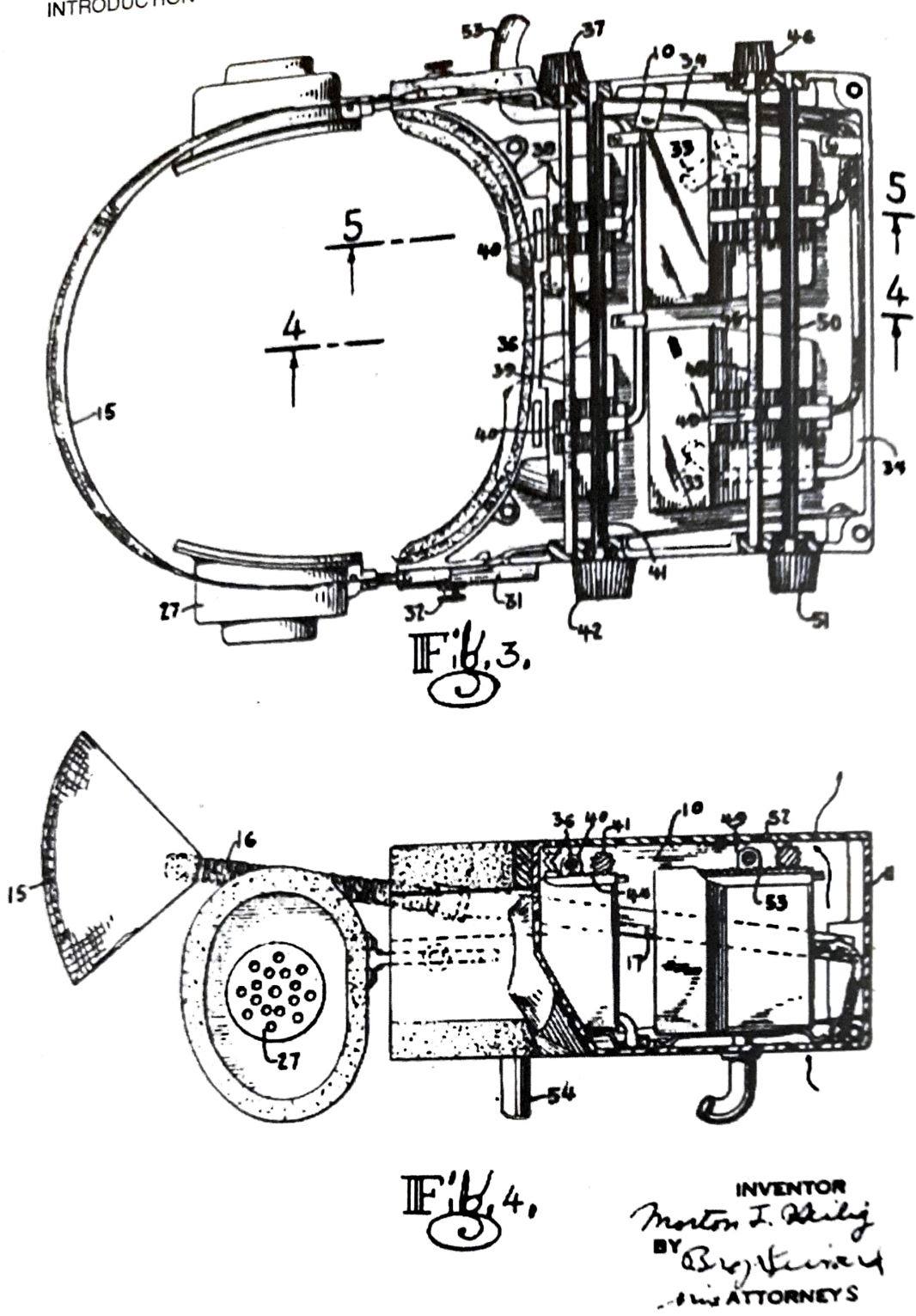


Fig. 1.4 Heilig's early head-mounted display patent. From Heilig [1960]. Reprinted by permission.

simulate two-dimensional continuous force fields associated with molecular docking forces [Batter and Brooks, 1971]. Later they simulated three-dimensional collision forces using a surplus robotic arm normally used in nuclear material handling. Most of today's haptic technology is based on miniature robotic arms.

The military was very eager to test the new digital simulators, since they hoped to replace very expensive analog ones. Flight simulators were hardware designed for a particular airplane model. When that airplane became obsolete, so did its simulator, and this was a constant drain of funds. If the simulation could be done in software

on a general-purpose platform, then a change in airplane models would only require software upgrades. The advantage seems obvious. Much research went on in the 1970s and early 1980s on flight helmets and modern simulators for the military, but much of this work was classified and was not published. This changed when funds for defense were cut and some researchers migrated to the civilian sector.

The National Aeronautics and Space Agency (NASA) was another agency of the American government interested in modern simulators. It needed simulations for astronaut training, as it was difficult or impossible to otherwise recreate conditions existing in outer space or on distant planets. In 1981, on a very small budget, NASA created the prototype of a liquid crystal display (LCD)-based HMD, which they named the Virtual Visual Environment Display (VIVED). NASA scientists simply disassembled commercially available Sony Watchman TVs and put the LCDs on special optics. These optics were needed to focus the image close to the eyes without effort. The majority of today's virtual reality head-mounted displays still use the same principle. NASA scientists then proceeded to create the first virtual reality system by incorporating a DEC PDP 11-40 host computer, a Picture System 2 graphics computer (from Evans and Sutherland), and a Polhemus noncontact tracker. The tracker was used to measure the user's head motion and transmit it to the PDP 11-40. The host computer then relayed these data to the graphics computer, which calculated new images displayed in stereo on the VIVED.

In 1985 the project was joined by Scott Fisher, who integrated a new kind of sensing glove into the simulation. The glove was developed earlier by Thomas Zimmerman and Jaron Lanier as a virtual programming interface for nonprogrammers. A photo of Fisher experimenting with the VIVED system is shown in Figure 1.5. By 1988 Fisher and Elizabeth Wenzel created the first hardware capable of manipulating up



Fig. 1.5 NASA VIVED prototype. Courtesy of Telepresence Research Inc.

to four 3D virtual sound sources. These are sounds that remain localized in space even when the user turns his or her head. This represented a very powerful addition to the simulation. The original VIVED project became VIEW (for Virtual Interface Environment Workstation) and the original software was ported to a newer Hewlett-Packard 9000, which had sufficient graphics performance to replace the wireframe rendering used in VIVED with more realistic flat-shaded surfaces.

With all the aforementioned technological developments, scientific exchange of information among the small group of specialists of the time followed. France was one of the first countries to organize a major international conference on the subject, held in Montpellier in March 1992. The name of this conference was Interfaces for Real and Virtual Worlds, and it drew hundreds of papers and many vendors. Later the same year the United States organized the first conference on Medicine Meets Virtual Reality. In San Diego, about 180 medical practitioners met with 60 scientists and engineers to discuss the great potential of virtual reality as a tool for medicine. In September 1993 the world's largest professional society, the Institute of Electrical and Electronics Engineers (IEEE), organized its first VR conference in Seattle. Virtual reality had become part of the mainstream scientific and engineering community.

1.3 EARLY COMMERCIAL VR TECHNOLOGY

The first company to sell VR products was VPL Inc., headed by Jaron Lanier. Until its demise in 1992 this company produced the first sensing glove, called the DataGlove (Figure 1.6a) [VPL, 1987]. The standard interfaces of the time (and still today) were the keyboard and the mouse. Compared to these, the VPL DataGlove represented a quantum improvement in the natural way one could interact with computers. Its fiber-optic sensors allowed computers to measure finger and thumb bending, and thus interaction was possible through gestures. Its drawbacks were high price (thousands of dollars), lack of tactile feedback, and difficulty in accommodating different hand sizes.

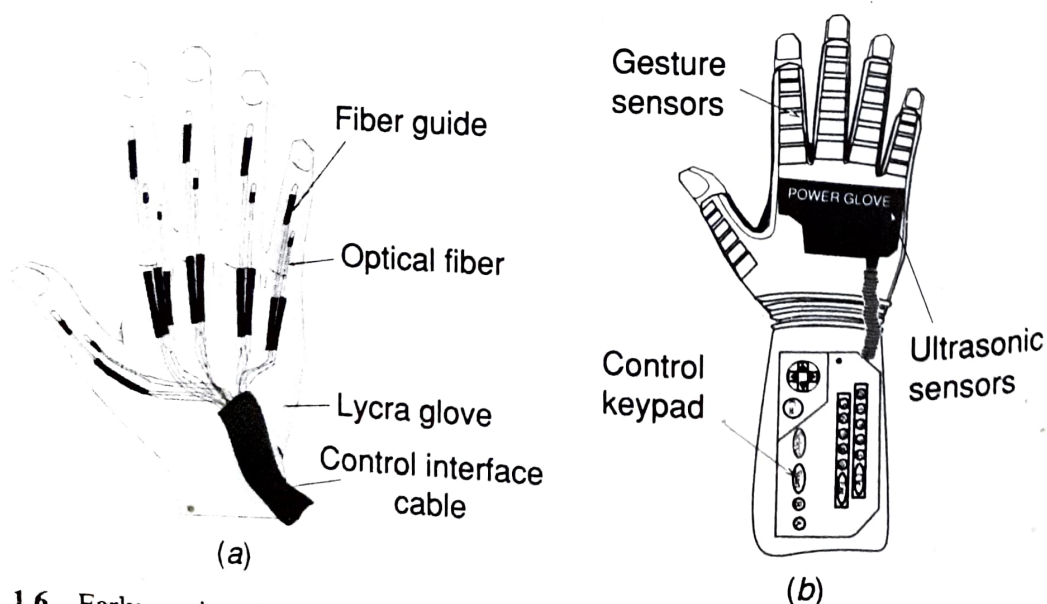


Fig. 1.6 Early sensing glove technology: (a) the VPL DataGlove; (b) the PowerGlove. From Burdea [1993]. Reprinted by permission.

Shortly after the appearance of the VPL DataGlove, the game company Nintendo introduced the much cheaper PowerGlove, shown in Figure 1.6b [Burdea, 1993]. It used ultrasonic sensors to measure wrist position relative to the PC screen and conductive ink flex sensors to measure finger bending. In 1989 almost one million such new game consoles were sold in a consumer frenzy that was later repeated with the introduction of Sony Play Station. The downfall of the PowerGlove was lack of sufficient games that used it, such that by 1993 its production had stopped.

The first commercial head-mounted displays, called EyePhones, were introduced by VPL in the late 1980s. These HMDs used LCD displays to produce a stereo image, but at extremely low resolution (360×240 pixels), such that virtual scenes appeared blurred. Other drawbacks were high price (\$11,000 each) and large weight (2.4 kg).

Researchers now had an initial set of specialized hardware with which to start developing applications. However, they first had to solve various integration issues as well as develop most of the required software from scratch. The idea of a turnkey VR system originated with VPL as well. Its RB2 Model 2 offered a rack assembly housing the EyePhone HMD interface, the VPL DataGlove Model 2 electronic unit, a spatial tracking unit for the HMD, a design and control workstation, as well as connections to an SGI 4D/310 VGX graphics renderer and to an optional 3D sound system.

The next step in integration was to shrink each of these components and put them on a board in a single desk-side cabinet. In early 1991 a company in the United Kingdom, Division Ltd., introduced the first integrated commercial VR workstation. It was called Vision and was followed by the more powerful Provision 100 [Grimsdale, 1992], which is illustrated in Figure 1.7. The Provision 100 parallel architecture had



Fig. 1.7 The Provision 100 VR workstation. Courtesy of Division Ltd.

multiple processors, stereo display on an HMD, 3D sound, hand tracking, and gesture recognition. The architecture also had an input/output (I/O) card and was scalable, allowing additional I/O processors to be added. Its i860 dedicated geometry processor with a custom polygon accelerator provided 35,000 Gouraud-shaded and Z-buffered polygons per second. This was a clear improvement over the speed of the HP 9000 computer used in NASA's VIEW system, but it came at a high price (\$70,000).

Although turnkey hardware systems appeared in early 1990s, the VR software development and debugging time continued to be a problem. In 1992, the small U.S. company Sense8 Co. developed the first version of its WorldToolKit (WTK) [Sense8 Co., 1992], a library of C functions written specifically for VR applications. With this toolkit developing VR software became more like a science, and debugging time was reduced significantly.

Another popular toolkit of the 1990s was the Virtual Reality Toolkit (VRT3), developed in the United Kingdom by Dimension International (the company later became Superscape PLC) [Dimension International, 1993]. Unlike WTK, VRT3 was designed to run on multiple computing platforms without the need for (then) pricey graphics accelerators. Also, unlike WTK, which is text-based, VRT3 used graphical programming through menus and icons. This made programming easier to learn, but less rich in functionality, owing to the limited menu of available functions and the limited number of supported interfaces.

1.5 THE FIVE CLASSIC COMPONENTS OF A VR SYSTEM

The discussion in this book will focus on the five classic components of a VR system, as depicted in Figure 1.11 [Burdea and Coiffet, 1993]. Chapters 2 and 3 concentrate

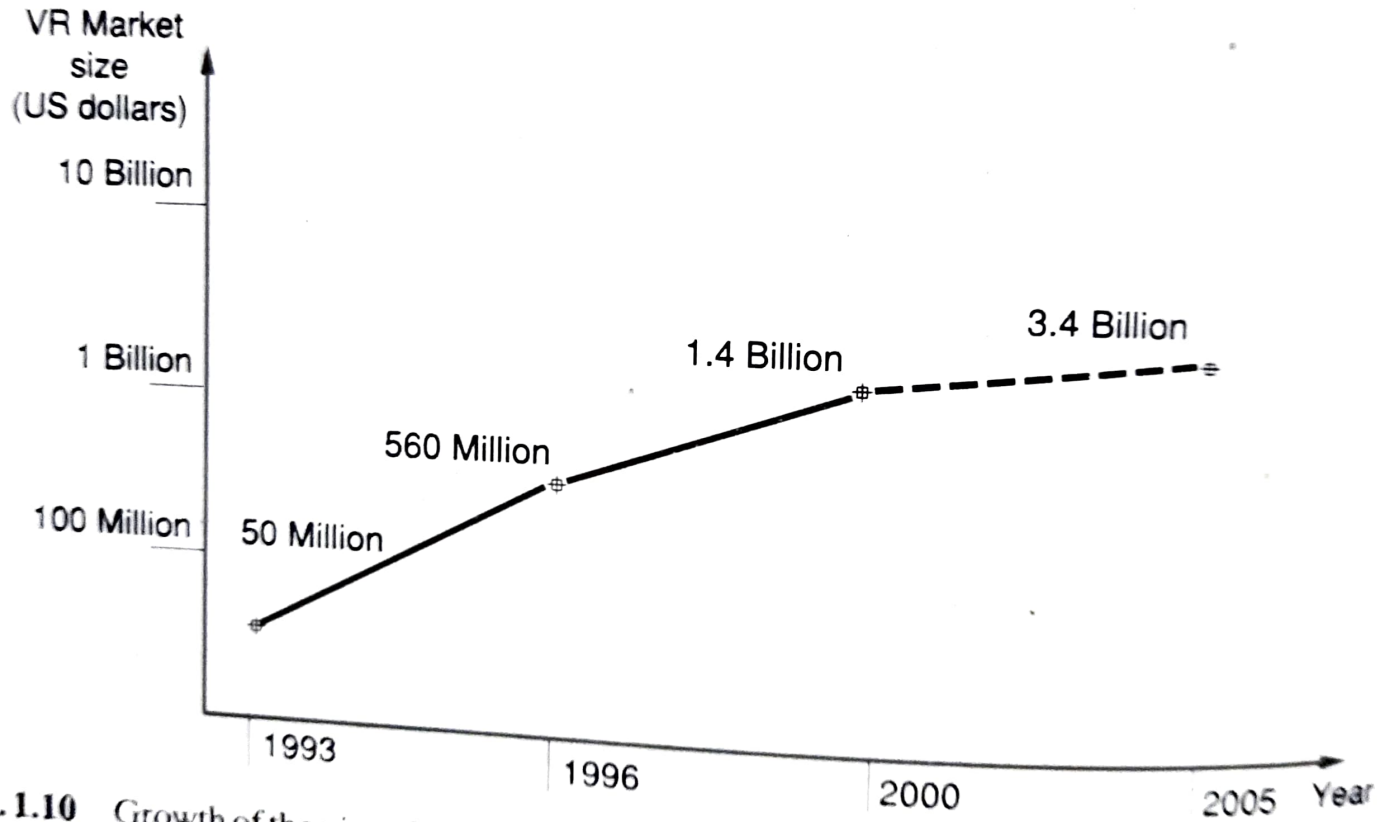


Fig. 1.10 Growth of the virtual reality industry since 1993. Based on Donovan [1993], Delaney [1997], and Machover [2000].

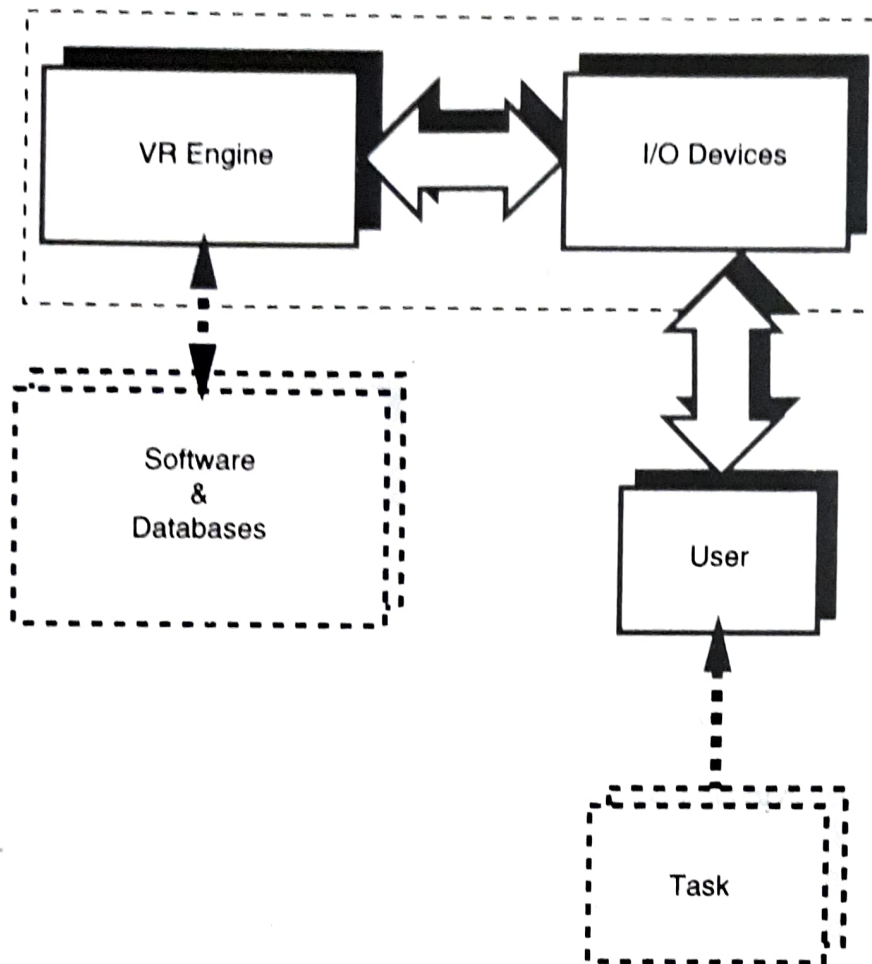


Fig. 1.11 The five classic components of a VR system. From Burdea and Coiffet [1993]. © Editions Hermes. Reprinted by permission.

on the very important I/O devices used either for user input (such as trackers, gloves, or mice) or output (such as HMDs, large-volume displays, force feedback robotic arms, etc.). The special-purpose computer architecture designed to match the high I/O and computation demands of real-time VR simulations is presented in Chapter 4. Chapter 5 deals with software for virtual object modeling, involving its geometry, texture, intelligent behavior, and physical modeling of hardness, inertia, surface plasticity, etc. Chapter 6 reviews a number of powerful programming packages, such as WorldToolKit and Java 3D, designed to help the virtual reality application developer. In Chapter 7 we analyze several human factors issues affecting simulation efficiency as well as user comfort and safety. Chapter 8 discusses traditional VR applications. These integrate all the aspects discussed in earlier chapters and make VR a tool for solving various practical problems in medical care, education, arts, entertainment, and the military. A look at emerging VR applications in manufacturing, robotics, and information visualization makes up the subject of Chapter 9, which concludes the book.

1.6 REVIEW QUESTIONS

1. What is virtual reality?
2. How does virtual reality differ from augmented reality and telepresence?

3. What are commonalities and differences between virtual reality and 3D computer graphics?
4. What was Heilig's role in the development of VR?
5. What was NASA's role in early VR developments and why was it interested?
6. What were the first commercial VR products?
7. What happened with the VR industry in the 1990s?
8. What are the five classic components of a VR system?